Supplementary Material: Literature Selection and Comparative Evidence

This supplement accompanies *Wheeled Mobile Robot Autonomy Under Degradation: A Survey of Methods and Evaluation*. It provides the selection and assessment details summarized in Section III, selected numerical comparisons discussed in the technical synthesis, and the tables of representative implementations referenced in Sections IV–VIII. Counts describe the selected literature and do not estimate research-topic prevalence across the field.

S1. Search Coverage and Selection

The initial search emphasized 2021–2026 publications across 17 journals and 102 journal–year combinations. It yielded 58,583 merged records, of which 43,086 had sufficient metadata for title/abstract screening. Language-model-assisted screening used preset topic allocations and stopped after 45 acceptances, leaving 43,041 metadata-complete records unassessed. Selection emphasized mechanisms relevant to compact wheeled platforms. The screening order and complete rejection history are unavailable. These missing records and the stopping rule limit the representativeness of the sample.

The archived screening decisions of the later candidate queues comprise 220 records, of which 150 accept, 40 reject, and 30 defer a candidate. Every record names Codex as reviewer and execution host. All but four state one of three title-and-abstract relevance policies. The read scope is recorded for 85 records: title and abstract for 74, title only for 7, metadata only for 3, and the abstract with the authors' platform description for one. Hard-gate results are recorded for 119 records, of which 94 passed. Gate observations are recorded for 43 more. An identity check of DOI, venue, year, publication type, and main-conference status is recorded for 57 records. No record states the language model or its version. The initial screening left only the aggregate counts given above. The public data list the follow-up screening of 79 candidates with reviewer and read scope.

Additional coverage comprised 5 complementary publications, a 60-work thematic sample, 15 studies selected for the four integration questions developed in the survey, 20 IEEE publications concerning motion priors, recovery, shared-state diagnosis, and reproducibility, a further 48 targeted works, 36 recent IEEE publications, and 16 coverage additions. The 48-work stage covers degeneracy and map validity, constrained motion and terrain risk, multimodal navigation, communication-aware collaboration, and recovery or online learning. The recent IEEE stage selected six 2025–2026 IEEE journal or ICRA papers for each of the six themes after an exact-DOI check against the works already selected. Table S1 lists all stages with their counts. The thematic sample contained 45 journal articles and 15 conference papers, with ten works under each of six themes: perception/localization, mapping/exploration, planning/control, multi-robot coordination, semantic navigation, and reliability/deployment. Searches combined technical, scenario, and mechanism terms with citation follow-up. Screening-based selection was completed by November 3, 2026, leaving coverage of the 2026 calendar year incomplete.

TABLE S1
Selection stages of the 245 publications. Each row adds publications to the earlier stages. IEEE counts the works published by IEEE.

Selection	Basis	Works	IEEE
Initial sample	Capped journal selection, 2021–2026	45	38
Complementary publications	Four technical studies and one context review	5	3
Thematic sample	Ten works under each of six themes	60	42
Integration questions	Four questions on combined navigation functions	15	8
Targeted IEEE coverage	Motion priors, recovery, shared-state diagnosis, and reproducibility	20	20
Further targeted coverage	Map validity, terrain risk, multimodal navigation, collaboration, and adaptation	48	21
Recent IEEE coverage	Six 2025–2026 IEEE journal or ICRA papers per theme	36	36
Coverage additions	2021–2024 works named in an internal review of an earlier draft	16	9
Total		**245**	**177**

Table S2 gives venue type and publisher by stage. The targeted and recent IEEE stages searched IEEE venues only, which makes all 56 of their works IEEE publications. The records of these stages state no reason for the restriction. Outside them, IEEE publications form 64% of the works. Without the two stages, the headline share of Section S6 is 33% instead of 33% (Table S11).

The final stage, on November 4, 2026, added 16 works from 2021–2024 that an internal review of an earlier draft named as missing. They comprise foundation-model and open-vocabulary navigation [1]–[5], uncertainty-aware terrain risk [6], [7], outlier-robust multi-robot SLAM [8], [9], DARPA Subterranean Challenge team systems [10]–[12], and datasets for degraded SLAM, off-road perception and dynamics, and embodied navigation [13]–[16]. They were not identified by a systematic search. The review also named VLMaps, whose 2025 journal extension is already in the corpus [17], NeBula, which exists only as a preprint, and further works that this stage did not acquire. Without these additions, 77 of the 228 purposively selected technical publications (34%) report a task

TABLE S2
VENUE TYPE AND PUBLISHER OF THE SELECTED WORKS BY SELECTION STAGE. THE PUBLISHER FOLLOWS THE DOI PREFIX, OR THE OFFICIAL URL HOST FOR WORKS WITHOUT A DOI.

Stage	Works	Journal	Conf.	IEEE	Elsevier	Wiley	SAGE	Springer	PMLR	Other	IEEE %
Initial sample	45	45	0	38	3	3	1	0	0	0	84
Complementary publications	5	5	0	3	0	0	1	0	0	1	60
Thematic sample	60	45	15	42	6	1	3	1	2	5	70
Integration questions	15	13	2	8	1	1	1	2	2	0	53
Targeted IEEE coverage	20	17	3	20	0	0	0	0	0	0	100
Further targeted coverage	48	36	12	21	2	7	6	2	10	0	44
Recent IEEE coverage	36	34	2	36	0	0	0	0	0	0	100
Coverage additions	16	6	10	9	0	0	0	0	2	5	56
All stages	245	201	44	177	12	12	12	5	16	11	72

Other: AAAS, Robotics: Science and Systems, NeurIPS Datasets and Benchmarks, Field Robotics and Tsinghua University Press. The context review is included in its stage.

outcome under a named degradation condition in the same experiment, against 81 of 244 (33%) with them.

A. Journal and Conference Sources

The journal searches covered Automatica, Autonomous Robots, Control Engineering Practice, IEEE Robotics and Automation Letters, IEEE Transactions on Automatic Control, IEEE Transactions on Automation Science and Engineering, IEEE Transactions on Cybernetics, IEEE Transactions on Image Processing, IEEE Transactions on Intelligent Transportation Systems, IEEE Transactions on Intelligent Vehicles, IEEE Transactions on Robotics, Information Fusion, Journal of Field Robotics, Pattern Recognition, Robotics and Autonomous Systems, Science Robotics, and The International Journal of Robotics Research. The 2025–2026 searches covered 34 journal–year combinations. Completion of a search provider's enumeration does not establish omission-free retrieval of the publisher's catalog.

Formal CoRL volumes for 2024 and 2025 and RSS indexes for 2024–2026 were consulted. ICRA/IROS papers were assessed individually, without exhaustive enumeration of their proceedings. Complete enumeration of a formal CoRL 2026 volume was not established. Overlapping results from different search sources are not independent screening denominators.

S2. ELIGIBILITY AND PUBLICATION SCOPE

Eligible publications include journal articles and main-conference papers concerning wheeled mobile robots or autonomy methods relevant to the navigation problems examined in the survey. Preprints and established conference-to-journal extensions are counted as one work, identified by title, authors, DOI or proceedings identifier, venue, and year. Publication-year statistics use formal volume years, whereas bibliography entries retain publishers' citation years, which can instead refer to online publication.

The selected literature contains 201 journal articles and 44 conference papers: 244 technical studies and one context review on learned navigation [25]. All corpus works are from 2021–2026, with two assessed studies from before 2021 left uncited. The 25 background references outside the corpus comprise representative estimators, related surveys, and foundational sources that attribute the definitions and equations

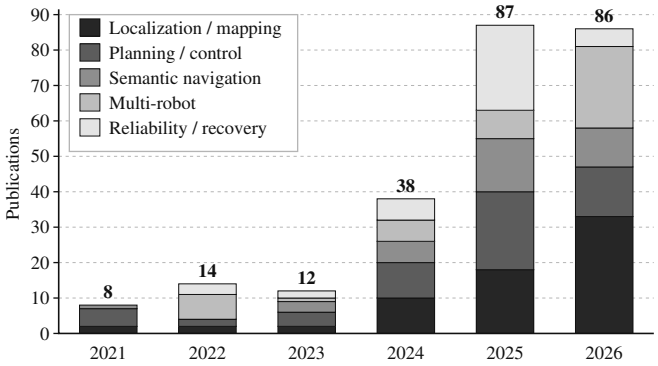


Fig. S1. Selected publications by year and principal navigation function. Each publication appears once. The distribution reflects purposive sampling and topic quotas.

of Section II: the kinematic classification of wheeled robots, the dynamic-window, timed-elastic-band, barrier-function, and Nav2 papers, a SLAM formulation, and a GNSS integrity review.

Of the 244 technical works, 172 were evaluated on target wheeled robots or their data, 12 on road vehicles (mark R), 13 on legged or tracked ground robots (G), and 47 on aerial, aquatic, handheld, or mixed carriers or without a specified platform (N). The implementation tables mark the last three groups. Mixed datasets that include a wheeled carrier stay in group N. A sensitivity row adds them to the wheeled subset. The non-wheeled works inform mechanism comparisons when the transfer criteria of Section III hold, with platform differences considered when assessing transfer. The target wheeled works also vary in drivetrain, sensing, and computational resources.

For the technical synthesis, publications are grouped by their principal contribution to localization/mapping, traversability/planning/control, semantic navigation, multi-robot coordination, or fault management. A publication may also inform other areas, although it appears only once in Fig. S1. These five functional groups differ from the six themes used to allocate the thematic sample.

TABLE S3
SELECTED RESULTS AND WITHIN-STUDY COMPARISONS. TABLE AND FIGURE NUMBERS IN THE FIRST COLUMN REFER TO THE CITED PAPERS. EACH ROW RETAINS ITS OWN METRIC AND PROTOCOL.

Study / source	Protocol and measure	Reported result	Interpretation
LODESTAR [18] Tables III and V	SubT-MRS Long Corridor replay: absolute pose error (APE). Desktop computation averaged across benchmark recordings.	APE 1.386 m versus 3.113 m for FAST-LIO2; benchmark-average computation 28.69 versus 21.04 ms.	Selective anchoring and point selection improve this sequence at additional computation cost; no LiDAR-blackout or closed-loop test.
FIT-SLAM 2 [19] Tables 2 and 4	Exploration with simulator poses and common control; completion time, plus Fisher-information timing analysis.	Cave: 880 versus 1118 s for FAEL; Corridor: 1809 versus 1706 s. Planning: 20.8 ms without and 63.0 ms with Fisher information.	The ordering changes with the scene. Planner benchmarks and the SLAM/localizability ablation answer different questions.
Safe Horizon MPC [20] Table 2	100 Gaussian eight-pedestrian simulations; common solver/cost and matched joint-risk specification 0.05; duration mean (SD).	16.0 (1.9) s versus 18.4 (2.4) s for CC-MPC. Maximum estimated plan collision probabilities: 0.0092 versus 0.0019.	Faster traversal accompanies less conservative plan risk under a known motion distribution; these are not empirical mission collision rates.
LeLaN [21] Tables 2 and 3	Visible-target navigation: 150 trials, 28 objects, success within 0.2 m; separate 15-trial obstacle ablation.	Goal success 0.89 versus 0.67 for OWL-v2 + ZoeDepth. With obstacles: arrival 0.60 with versus 0.13 without collision training.	Target arrival and obstacle handling are separate evaluations; visible-target success does not establish open-world search or a collision-risk bound.
TAPM-IVP [22] Table 2	Scene 1 simulation; two robots, ten independent trials per method, common local planner/tracker.	Success 90% for each robot versus 40% and 30% in the two-robot ablation without partitioning and greedy subgoal selection.	The ablation supports the combined design, not either module alone. Rates are per robot; hardware demonstrations use two Ackermann vehicles.
GP-based recovery [23] Fig. 6 and Section V	Ten simulated runs per goal/world; same planner with and without recovery. Counts read from the plotted rates.	Dense 1 m doorways, goal 0: 8/10 versus 3/10 successes. Sparse 2 m doorways, goal 0: 10/10 for both.	Recovery benefit depends on the failure setting. The physical demonstrations do not provide a comparable success-rate estimate.
Home deployment [24] Table 3	Ten-home field deployment; three assistive-service types, including requested but unperformed interventions.	1130 successful, 177 failed, and 232 unperformed interventions out of 1539 requests. Reported success: 86.46% of 1307 attempts.	The success denominator excludes unperformed requests; availability and assistance conditions remain relevant to service-level interpretation.

S3. ASSESSMENT AND COMPARISON

Comparison attributes include the technical mechanism, platform, sensing, computation, experimental setting, baselines, measured outputs, and limitations. For 23 publications, assessment covered the full paper and numerical tables, whereas for 222 it was limited to selected method, evaluation, and conclusion passages or existing extracts. Of these 222 publications, 87 assessments were based on earlier extractions and 135 on targeted readings. These assessment levels do not provide uniform whole-paper verification across the selected literature. Each value of the evidence coding that is read from a paper carries a source location, either a physical PDF page or a review-card block that, on a light card, cites a page. Values that rest on an earlier extraction were located on the PDF page. Platform group, drivetrain, strongest evidence setting, integrity assessment, and computation placement follow recorded card fields and have no location of their own.

The 48, 36, and 16 works of the last three stages received targeted readings with page-located excerpts matched to both the transcription and PDF text. Their reading records retain mechanism and protocol conditions. Numbers that the survey reports from these works were rechecked in the PDF text, without rendered-table audits. Four citation entries were rewritten from the publisher's citation file, which is kept unchanged, including an early-access year correction and the replacement of an unparsable citation key. One further work has a title that differs between its publisher page and its PDF. The public data record these citation changes separately from the reading depth.

Experimental settings distinguish hardware control, online estimation, recorded robot data, dataset replay, simulation, and numerical studies. Dataset and review publications are identified separately. Tables S4–S8 summarize mechanisms and inputs alongside experimental settings, evaluated quantities, and computation location. Interpretation accounts for external pose, remote computation, low speed, and staged scenes where applicable. Complete-pipeline timing and placement require information beyond processor identity.

Quantitative comparisons retain the reported metrics, baselines, and evaluation conditions. Effect sizes are not pooled across heterogeneous datasets, platforms, or trial units. The evaluation table of the survey counts attributes over the 244 technical publications, excluding the context review. It groups overlapping attributes by measurement level, degradation evidence, same-experiment combination, evidence setting, assessment, and platform group (Section S6). A study may report component and task measures during a degradation test. Purposive selection, uneven assessment depth, and heterogeneous protocols limit the generality of comparative conclusions.

S4. SELECTED RESULTS AND WITHIN-STUDY COMPARISONS

Table S3 retains the measures and experimental units of seven selected examples. Numerical results were checked against the cited tables or plots. Each comparison concerns the stated scene, baseline, and information available to the

method. The rows therefore do not form a cross-study ranking. The examples include changes in performance ordering, added computation, combined-module ablations, and success denominators to clarify the scope of the conclusions drawn in the survey.

S5. Representative Implementations by Navigation Function

Tables S4–S8 list representative implementations for the five functional sections of the survey. Each row reports the mechanism and its inputs, the experimental setting with the evaluated quantities, and the platform with the computation placement stated by the authors. The rows were chosen to cover the response families discussed in each section, and they are not a ranking or a complete list of the selected works.

S6. Evidence Coding

The codebook, the per-work codes with their source locations, the second coding, the adjudication log, and a list of all 245 works with selection stage, principal section, venue, publisher, and platform group are released as public data with the survey. The coding unit is the publication, whose attributes summarize every experiment it reports. A combined attribute counts only when both parts come from one experiment, identified by its trial set, table, or figure. Degradation evidence is coded per source in three classes. D1 marks a condition that the authors inject, remove, or vary and report for the same system at a second level, or with and without the evaluated response under the same condition. D2 marks a condition selected for its difficulty without manipulation, including a manipulated condition reported at one level only. D3 marks a condition that occurred during the experiments and is reported without being designed in. A condition mentioned only as motivation is not coded. Moving agents count when the study treats them as a disturbance or varies them. Model error counts only when wrong or out-of-distribution outputs are evaluated.

Nine sources form five groups and violate six assumptions (Fig. 1 of the main paper). A work occupies one matrix cell for every source it addresses, always in the column of its principal section, and counts once in every pair of the functions it codes. Platform groups follow the platform class of the evaluation. The drivetrain follows the class of the target wheeled works. A drivetrain-dependent effect is coded only when a paper reports an effect of slip, turning radius, lateral motion, reversing, or footprint that depends on the drive, which 10 works do. Assessment depth distinguishes full readings, earlier extractions, and targeted readings, as in Section S3.

Tables S9–S11 give the column percentages of the evidence attributes, the works of every matrix cell, and the headline share under alternative definitions and subsets with four robustness findings fixed before primary coding. Findings RF1 (localization has the lowest headline share) and RF4 (task outcomes in at most 5% of localization works) hold under every definition and in every stage, depth, and quota subset, with a tie among full readings. RF1 fails only among works from 2026. RF4 fails only in one quota theme. RF2 (fault management has the highest share) and RF3 (task outcomes in at least 90% of planning and semantic works) fail under the final codes. Planning has the highest share. Task outcomes appear in 44 of 56 planning and 33 of 36 semantic works. The card-level coding of the earlier draft had supported RF2.

S7. Language-Model Second Coding

A stratified random sample of 40 technical works, eight per principal section, was drawn with a seed fixed in the codebook before primary coding. Within each section, the draw takes two full readings and splits the other six between earlier extractions and targeted readings by their section shares. The draw was repeated until every selection stage appeared. A separate language-model session, configured for a different model, coded the thirteen double-coded attributes from the codebook, the sample list, and the review cards and transcriptions of the sampled works. It had no access to the primary codes, which another session wrote for all 245 works and froze by hash. The public coding record names the model configured for each pass and gives the timeline of the two passes. Neither session recorded the model that actually ran.

Table S12 reports percent agreement and unweighted Cohen's kappa before adjudication. Both coders are language-model sessions that read review cards themselves prepared with language-model tools. The statistics therefore describe how consistently two such coders apply the codebook to shared evidence and are not human inter-rater reliability. Among the attributes, kappa is lowest for the drivetrain-dependent effect, which the second coder assigned to four works and the primary coder to one, and for the nominal control. The maximum of 1.00 belongs to incidental sources and rests on the same 3 of 360 source flags set by both coders. A third language-model session with access to both codings and the sources re-read every one of the 112 differing values at both locators and logged a reason and the deciding location. It kept the primary value 58 times, adopted the second value 46 times, and revised both 8 times. Codebook version 1.1 clarified the two rules that had caused disagreement in at least three works, by limiting the response family to its primary value and by narrowing the drivetrain-dependent effect. That effect now excludes differences between wheeled and other locomotion, limits that apply to every wheel arrangement, and kinematic models without a reported geometry-dependent result. A recheck of these rules across the corpus changed six further values. After an internal re-review, a further recheck searched all works for drivetrain-dependent effects that the primary codes might have missed, through a keyword screen of every review card and transcription and a reading of each candidate passage on its PDF page. It added 4, for 10 works in all. The same recheck read the PDF pages of the 19 works coded at the physical evidence level without a physical task outcome under a named condition. It kept the level where the physical run states a result under that condition, lowered it for 2 works, and corrected the source locations of 2 works. For 5 of the kept works, the text or a table of the physical run reports a task outcome under that condition. The recheck coded these combinations as physical. A consistency check of the final codes now requires a stated result for the physical

TABLE S4
LOCALIZATION AND MAPPING: REPRESENTATIVE MECHANISMS, SENSING, AND EVALUATION SETTINGS.

Work	Mechanism and inputs	Evaluation setting *Evaluated quantities*	Platform *Computation*
Uncertainty-aware localization [26]	Uncertainty-weighted prior-map matching with LIO and gated GNSS. *Inputs:* 3D LiDAR, IMU, GNSS	Acquired data *Pose error and covariance estimates*	wheeled (drive unspecified) *Unspecified*
LODESTAR [18]^{N}	Selective historical anchoring and point selection under degeneracy. *Inputs:* 3D LiDAR, IMU	Replay *Absolute pose error and runtime*	mixed *Desktop evaluation*
ToF-Wheel [27]	Directional process-covariance inflation from geometric corrections. *Inputs:* time-of-flight camera and wheels; LiDAR replaces depth in replay	Acquired data, Replay *Trajectory error across covariance variants*	differential drive *Onboard*
DURAL [28]	Region- and degeneracy-dependent sensor fusion. *Inputs:* LiDAR, IMU, UWB, wheel encoders	Acquired data, Sim. *Localization error across fusion modes*	wheeled (drive unspecified) *Unspecified*
MM-LINS [29]^{R}	Map hibernation, reinitialization, and pose-graph registration. *Inputs:* LiDAR/IMU from handheld and vehicle rigs	Acquired data, Replay *Trajectory error and map registration*	road vehicle *Unspecified*
SE2-LIO [30]	Near-planar pose with stochastic out-of-plane perturbations. *Inputs:* 3D LiDAR, IMU	Online estimate, Acquired data *Pose error; slope-model boundary*	wheeled (drive unspecified) *Onboard*
ProbPer-LiLo [31]^{N}	Semantic-cluster persistence estimation and map filtering. *Inputs:* 3D LiDAR, semantic labels	Acquired data, Replay *Registration and map persistence*	mixed *Unspecified*
DMSA [32]^{N}	Joint grid-distribution fitting over multiple scans. *Inputs:* 3D LiDAR, IMU	Acquired data, Replay *Pose error, runtime, sensor compatibility*	mixed *Desktop evaluation*
Switch-SLAM [33]^{N}	Degeneracy-triggered introduction of visual constraints. *Inputs:* LiDAR, camera, IMU	Replay, Sim. *Trajectory error under modality failure*	mixed *Desktop evaluation*
M2UD [34]	Uneven-terrain localization and mapping benchmark. *Inputs:* LiDAR, RGB-D, IMU, GNSS; mixed reference systems	Dataset, Acquired data *Localization/mapping error and runtime*	wheeled (drive unspecified) *Unspecified*
DR-ICP [35]^{N}	Directional feature selection and weighting. *Inputs:* LiDAR; capture IMU not integrated	Acquired data, Replay, Sim. *Tunnel pose error; pure-LiDAR scope*	mixed *Desktop evaluation*
Decoupled registration [36]^{N}	Subspace decoupling and directional preconditioning. *Inputs:* LiDAR registration constraints	Acquired data, Replay, Numerical *Registration; missing-information boundary*	mixed *Desktop evaluation*
Semantic neural processes [37]^{R}	Semantic neural-process elevation and uncertainty. *Inputs:* LiDAR, image semantics, poses	Acquired data *Elevation error and uncertainty; calibration needed*	road vehicle *Desktop evaluation*

Superscripts mark evaluations on road vehicles (R), other ground robots (G), and non-ground, handheld, mixed, or unspecified carriers (N).

level. The headline count is 81 in the frozen primary codes and 81 after adjudication and the later rechecks.

S8. DERIVATION OF THE OPEN INTERFACES

A candidate interface is an unordered pair of distinct functions with one coupling source. Its works code both functions and address, name, or report the source. The rule, fixed in codebook version 1.0 before coding (commit 67872f4), keeps candidates with at least three works and sorts them by the number of works without direct joint wheeled evidence, then by safety relevance (planning or fault management in the pair), then by the number of target wheeled works. At equal direct evidence, this count ranks a candidate with more works above one with fewer, with candidate sizes shaped by the selection stages, including the integration-questions stage acquired for the interfaces. Table S13 gives the function-pair co-occurrence with evidence levels. Table S14 ranks the candidates at or above the lowest developed interface, with every interface cell, two sensitivity ranks, and the reason for each candidate that is not developed.

Each interface takes the rank of its best eligible cell. Interface 4 has rank 1 (LOC-MRC:COM), Interface 1 rank 4 (LOC-REL:PER), Interface 2 rank 10 (PLN-SEM:MOD), and Interface 3 rank 11 (LOC-PLN:PER) among 32 eligible candidates. The computation-limit cell of Interface 2 (PLN-SEM:RES) ranks 26, whereas the terrain cell of Interface 3

TABLE S5
TRAVERSABILITY, PLANNING, AND CONTROL: REPRESENTATIVE MECHANISMS, SENSING, AND EVALUATION SETTINGS.

Work	**Mechanism and inputs**	**Evaluation setting** *Evaluated quantities*	**Platform** *Computation*
LRAE [38]	Unknown-region safety classification and route optimization. *Inputs:* 3D LiDAR	Hardware, Sim. *Exploration time and travel cost*	skid steer *Unspecified*
CADNet [39]	Heading-dependent accessible-range prediction for TEB planning. *Inputs:* LiDAR history	Hardware, Replay, Sim. *Traversability prediction and navigation*	wheeled (drive unspecified) *Onboard*
GA-Nav [40]	Semantic terrain segmentation combined with elevation costs. *Inputs:* RGB images, 3D LiDAR	Hardware, Replay *Terrain segmentation and navigation success*	wheeled (drive unspecified) *Onboard*
Anisotropic exploration [41]	Direction-dependent quasi-static terrain costs. *Inputs:* LiDAR terrain map, FAST-LIO pose	Hardware, Sim. *Exploration, travel cost, body roll*	differential drive *Onboard*
FIT-SLAM2 [19]	Keyframe-linked frontiers and information-aware adaptive roadmaps. *Inputs:* RGB-D/LiDAR (simulation); stereo depth (hardware)	Hardware, Sim. *Exploration time, pose error, and runtime*	skid steer *Unspecified*
Vehicle–cloud MPC [42]	Switching between onboard linear and cloud nonlinear MPC. *Inputs:* externally measured poses	Hardware, Sim., Theory *Tracking and collision outcomes*	Ackermann *Mixed placement*
CECBF [43]	CBF constraints with circulation and hold logic. *Inputs:* 2D LiDAR, external robot pose	Hardware *Obstacle avoidance and escape from equilibria*	differential drive *Local offboard*
Safe Horizon MPC [20]	Scenario constraints on joint collision risk with deceleration fallback. *Inputs:* robot/pedestrian poses and velocity estimates	Hardware, Sim. *Task duration and planned collision risk*	skid steer *Unspecified*
Spatiotemporal-Tubes [44]	Adaptive spherical tubes for prescribed-time reach-avoid. *Inputs:* robot state, online unsafe-set geometry	Hardware, Sim., Numerical *Reach-avoid and prescribed-time behavior*	omnidirectional *Unspecified*
Perceive With Confidence [45]^{G}	Environment-level calibration of obstacle inflation for constrained planning. *Inputs:* RGB-D point clouds, VIO	Hardware, Sim. *Collision-free outcomes and calibration*	legged *Onboard*
Homotopy MPC [46]	Parallel optimization of homotopy alternatives. *Inputs:* externally tracked robot/pedestrian states	Hardware, Sim. *Dynamic avoidance; external pose*	wheeled (drive unspecified) *Unspecified*
Visual-geometric MPPI [47]	Expert-trained cost, speed, and novelty maps for MPPI. *Inputs:* RGB features, LiDAR geometry	Hardware, Acquired data *Interventions, speed, and terrain response*	wheeled (drive unspecified) *Unspecified*
SPARTA [48]	Angle-conditioned risk queried during planning. *Inputs:* prepared point cloud, LiDAR/IMU pose	Hardware, Sim. *Simulated tire risk; virtual risky hardware branch*	skid steer *Onboard*

Superscripts mark evaluations on road vehicles (R), other ground robots (G), and non-ground, handheld, mixed, or unspecified carriers (N).

(LOC-PLN:TER) has 2 works, below the eligibility threshold. The four interfaces therefore do not form the top of the ranking. The main paper presents them as an agenda proposed by the survey. Five of the 6 undeveloped candidates above Interface 3 are merged into Interfaces 4 or 2, whose comparisons vary their sources. The sixth, actuator faults within one tracking controller, is named as a further gap. The alternatives named by the internal review either rank below Interface 3 (map change forcing plan re-validation, perception latency, and energy or computation limits on planning) or fall outside the function pairs (human assistance, coded in 10 works). Removing the integration-questions stage, which was acquired for the interfaces, gives ranks 4, 8, 9, and 1 for Interfaces 1–4. Sorted by gap share (Rank B), Interfaces 1–4 rank 5, 26, 18, and 12, below the undeveloped actuator-fault cell PLN-REL:ACT. The public data list the contributing works of every interface with platform group and evidence level. Interfaces 1–4 have 1, 6, 1, and 0 works with direct joint wheeled evidence.

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TABLE S6
SEMANTIC AND LANGUAGE-GUIDED NAVIGATION: REPRESENTATIVE MECHANISMS, SENSING, AND EVALUATION SETTINGS.

Work	**Mechanism and inputs**	**Evaluation setting** *Evaluated quantities*	**Platform** *Computation*
VLM-Social-Nav [49]	Cue-triggered VLM judgments used as social DWA costs. *Inputs:* RGB images, LiDAR	Hardware *Task success, collisions, and social behavior*	differential drive *Mixed placement*
Co-NavGPT [50]^{G}	VLM frontier allocation with geometric goal execution. *Inputs:* RGB-D, shared map, robot states	Hardware, Sim. *HM3D search; separate hardware demos*	legged *Onboard pose; offboard perception/planning*
OpenIN [51]	Carrier–object memory with visibility-dependent updates. *Inputs:* RGB-D, language/image queries, SLAM pose	Hardware, Sim. *Success across sequential object queries*	wheeled (drive unspecified) *Unspecified*
VLFM [52]^{G}	Image–language similarity projected into a frontier value map. *Inputs:* RGB-D, odometry	Hardware, Sim. *Goal success and path efficiency*	legged *Unspecified*
HOV-SG [53]^{G}	Hierarchical retrieval over floor, room, and object graphs. *Inputs:* RGB-D, LiDAR-derived poses	Hardware, Sim., Replay *Object retrieval and navigation success*	legged *Unspecified*
Uni-NaVid [54]^{G}	Multiscale video memory and discrete action prediction. *Inputs:* monocular RGB video, instruction	Hardware, Sim. *Goal success and path efficiency*	legged *Local offboard*
LeLaN [21]	Receding-horizon language-conditioned velocity prediction. *Inputs:* RGB image, text instruction	Hardware, Replay *Target arrival and action-generation time*	differential drive *Onboard*
VL-TGS [55]	VLM selection among generated trajectories with local DWA control. *Inputs:* LiDAR, RGB, velocity, GPS	Hardware, Acquired data *Outdoor traversal and model-response time*	wheeled (drive unspecified) *Mixed placement*
GSON [56]	Group-aware routing with predictive barrier-constrained control. *Inputs:* 2D LiDAR, RGB images	Hardware, Replay *Social disturbance, travel time, and jerk*	differential drive *Unspecified*
AVLMaps [17]	Multimodal spatial retrieval and goal APIs. *Inputs:* offline RGB-D/pose/audio map, instructions	Hardware, Sim. *Multimodal goals; prepared-map validity*	wheeled (drive unspecified) *Unspecified*
VLM-GroNav [57]	Interaction-grounded terrain costs and DWA. *Inputs:* RGB, LiDAR, proprioception, GPS, aerial images	Hardware *Outdoor task outcomes; GPS/cloud dependencies*	skid steer *Mixed placement*
SocialNav-SUB [58]	Social scene-understanding VQA benchmark. *Inputs:* recorded images, pedestrian annotations	Dataset *Human/rule agreement; no closed-loop control*	wheeled (drive unspecified) *Mixed placement*

Superscripts mark evaluations on road vehicles (R), other ground robots (G), and non-ground, handheld, mixed, or unspecified carriers (N).

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TABLE S7
MULTI-ROBOT COORDINATION: REPRESENTATIVE MECHANISMS, SENSING, AND EVALUATION SETTINGS.

Work	Mechanism and inputs	Evaluation setting *Evaluated quantities*	Platform *Computation*
ROAM [59]	One-hop consensus for semantic mapping and viewpoint planning. *Inputs:* RGB-D, LiDAR	Hardware, Sim. *Distributed mapping and exploration*	skid steer *Onboard*
Active graph exploration [60]^{N}	Submodular loop-edge selection after coverage planning. *Inputs:* prior topo-metric graph	Sim. *Simulated coverage and graph objective*	none *Desktop evaluation*
Range integrity [61]^{N}	Range-consistency localization-error reconstruction. *Inputs:* emulated ranges, reported poses	Online monitor (mixed reality), Sim. *Error reconstruction with emulated ranges*	aerial *Central ground station; emulated ranges*
Commerge [62]	Descriptor matching and scan selection before submap registration. *Inputs:* 3D LiDAR, LIO poses	Acquired data, Replay *Map-merging error and transmitted data*	wheeled (drive unspecified) *Desktop evaluation*
iMESA [63]^{N}	Incremental local factor graphs with shared-variable consensus. *Inputs:* odometry and inter-robot factors	Replay, Numerical *Estimation error with limited communication*	mixed *Desktop evaluation*
Collaborative map repair [64]^{R}	Residual-based fault isolation and correlation-aware map repair. *Inputs:* LiDAR, wheel odometry; GNSS initialization	Acquired data, Sim. *Pose/map error and fault isolation*	road vehicle *Unspecified*
TAPM-IVP [22]	Terrain-aware 3D Voronoi allocation and subgoal selection. *Inputs:* LiDAR, A-LOAM poses	Hardware, Sim. *Success, travel cost, and path duplication*	Ackermann *Onboard*
Failure-aware tracking [65]^{N}	Failure-dependent tracking modes and target-state fusion. *Inputs:* target range/bearing and inter-robot messages	Hardware, Sim. *Target-state error during failures*	aerial *Unspecified*
Cross-rank [66]^{G}	Intersection allocation from shared position histories. *Inputs:* LiDAR localization and low-bandwidth radio	Hardware, Sim. *Exploration under intermittent reception*	legged *Mixed placement*
CoVOR-SLAM [67]	Similarity-pose-graph fusion for monocular scale correction. *Inputs:* camera, inter-rover UWB ranges	Acquired data, Replay, Sim. *Pose error and monocular scale correction*	wheeled (drive unspecified) *Desktop evaluation*
Swarm-SLAM [68]	Sparse loop prioritization and decentralized inference. *Inputs:* local odometry, loop observations	Online estimate, Replay *Online collaborative estimation; team exchange*	skid steer *Onboard*
Planned synchronization [69]	Joint motion and observation scheduling. *Inputs:* relative marker observations, IMU	Hardware, Numerical *Formation and check-in cost; modeled non-idealities*	differential drive *Onboard*
Bandwidth-efficient SLAM [70]	Lossy keyframe selection and lossless coding. *Inputs:* visual/inertial keyframes	Online estimate, Replay *Bandwidth/accuracy tradeoff; central back end*	wheeled (drive unspecified) *Mixed placement*

Superscripts mark evaluations on road vehicles (R), other ground robots (G), and non-ground, handheld, mixed, or unspecified carriers (N).

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TABLE S8
RELIABILITY AND RECOVERY: REPRESENTATIVE MECHANISMS, SENSING, AND EVALUATION SETTINGS.

Work	Mechanism and inputs	Evaluation setting *Evaluated quantities*	Platform *Computation*
Visual NMPC recovery [71]	Tracking-loss stop and relocalization; restart after failure. *Inputs:* stereo vision, actuator feedback	Hardware *Tracking; stop/relocalization handling*	skid steer *Unspecified*
Set-based LPV diagnosis [72]^{R}	Zonotopic detection and fault-class separation. *Inputs:* yaw rate, odometry, sideslip, steering angle	Acquired data *Detectability and isolability under injected faults*	road vehicle *Desktop evaluation*
Reliable MCL [73]	Joint pose/reliability estimation with global relocalization. *Inputs:* 2D LiDAR, odometry or inertial navigation	Acquired data, Replay, Sim. *Reliability estimates and relocalization*	wheeled (drive unspecified) *Unspecified*
Long-term home deployment [24]	Long-term service, docking, and remote human supervision. *Inputs:* LiDAR, RGB-D, docking and home sensors	Hardware *Service outcomes and interruptions*	wheeled (drive unspecified) *Mixed placement*
KSFD [74]	Kinematic residuals for sensor-fault diagnosis. *Inputs:* inertial channels and an inertial-aided position substitute	Acquired data, Numerical *Fault-isolation residual patterns*	Ackermann *Desktop evaluation*
Behavior-tree recovery [75]^{N}	Behavior-tree supervision and failure-specific recovery. *Inputs:* terrain maps, contacts, planner status	Hardware, Sim. *Recovery on retained scenarios; human-rescue boundary*	mixed *Onboard*
Line-map integrity [76]	Prior-line-map localization with an error protection level. *Inputs:* monocular images, IMU, prior 3D lines	Acquired data, Replay, Sim. *Pose error and protection-level coverage*	wheeled (drive unspecified) *Desktop evaluation*
Event-triggered tracking [77]	Observer-based fault estimation and event-triggered tracking. *Inputs:* vehicle-posture outputs in the model	Hardware-in-the-loop, Numerical *Tracking error and update frequency*	Ackermann *Mixed placement*
Sentinel [78]	Action-consistency and VLM progress monitoring. *Inputs:* external RGB-D and mobile-base motion capture	Online estimate, Sim. *Failure detection and nominal alarms*	omnidirectional *Mixed placement*
RTEB [79]	Hybrid A* reinitialization and local smoothing. *Inputs:* RTK pose, voxel obstacle map	Hardware, Sim. *Planner recovery; TEB/NMPC comparisons*	wheeled (drive unspecified) *Onboard*
SCALE [80]	Affordance prediction and anti-novelty recovery. *Inputs:* RGB at deployment; offline robot data	Hardware *Visual recovery and intervention need*	skid steer *Onboard*
LiReN [81]	Online RL with resets, goals, and charging. *Inputs:* RGB policy; LiDAR/map, bumper for supervision	Hardware *Adaptation with operational resets and infrastructure*	differential drive *Mixed placement*

Superscripts mark evaluations on road vehicles (R), other ground robots (G), and non-ground, handheld, mixed, or unspecified carriers (N).

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TABLE S9
EVIDENCE ATTRIBUTES OF THE TECHNICAL WORKS BY PRINCIPAL SECTION. EACH PERCENTAGE REFERS TO THE WORKS OF ITS COLUMN. ATTRIBUTES OVERLAP.

Attribute	IV	V	VI	VII	VIII	All
Publications	67	56	36	45	40	244
Task outcome reported	1 (1%)	44 (79%)	33 (92%)	25 (56%)	20 (50%)	123 (50%)
Task outcome in simulation only	0 (0%)	19 (34%)	9 (25%)	10 (22%)	11 (28%)	49 (20%)
Physical task outcome	1 (1%)	25 (45%)	24 (67%)	15 (33%)	9 (22%)	74 (30%)
Degradation source addressed	65 (97%)	53 (95%)	24 (67%)	38 (84%)	39 (98%)	219 (90%)
Named condition (D1 or D2)	61 (91%)	46 (82%)	19 (53%)	27 (60%)	34 (85%)	187 (77%)
Injected or varied with a comparator (D1)	16 (24%)	15 (27%)	6 (17%)	13 (29%)	11 (28%)	61 (25%)
Selected condition only (D2)	45 (67%)	31 (55%)	13 (36%)	14 (31%)	23 (58%)	126 (52%)
Incidental condition only (D3)	1 (1%)	1 (2%)	2 (6%)	4 (9%)	2 (5%)	10 (4%)
Named condition with a task outcome in the same experiment	1 (1%)	36 (64%)	17 (47%)	11 (24%)	16 (40%)	81 (33%)
D1 condition with a task outcome in the same experiment	0 (0%)	12 (21%)	6 (17%)	5 (11%)	4 (10%)	27 (11%)
Same-experiment combination on a physical robot	1 (1%)	20 (36%)	13 (36%)	2 (4%)	7 (18%)	43 (18%)
Direct joint wheeled evidence	1 (1%)	17 (30%)	12 (33%)	1 (2%)	6 (15%)	37 (15%)
Named condition, physical closed loop	1 (1%)	24 (43%)	14 (39%)	5 (11%)	11 (28%)	55 (23%)
Named condition, online hardware or recorded data	46 (69%)	3 (5%)	0 (0%)	6 (13%)	11 (28%)	66 (27%)
Named condition, replay, simulation, numerical model or resource	14 (21%)	19 (34%)	5 (14%)	16 (36%)	12 (30%)	66 (27%)
Integrity or failure-risk assessment	3 (4%)	1 (2%)	0 (0%)	1 (2%)	9 (22%)	14 (6%)
Computation placement reported	45 (67%)	31 (55%)	18 (50%)	33 (73%)	28 (70%)	155 (64%)
Target wheeled robot	30 (45%)	50 (89%)	30 (83%)	32 (71%)	30 (75%)	172 (70%)
Road vehicle	7 (10%)	1 (2%)	0 (0%)	0 (0%)	4 (10%)	12 (5%)
Other ground robot	1 (1%)	4 (7%)	5 (14%)	3 (7%)	0 (0%)	13 (5%)
Non-ground, handheld, mixed-carrier or no platform	29 (43%)	1 (2%)	1 (3%)	10 (22%)	6 (15%)	47 (19%)
Drivetrain-dependent effect reported	3 (4%)	3 (5%)	0 (0%)	0 (0%)	4 (10%)	10 (4%)

TABLE S10
WORKS THAT ADDRESS EACH DEGRADATION SOURCE, BY PRINCIPAL SECTION. A WORK APPEARS IN THE ROW OF EVERY SOURCE IT ADDRESSES AND ALWAYS IN THE COLUMN OF ITS PRINCIPAL SECTION.

Source (assumption)	IV	V	VI	VII	VIII
Perceptual degradation (A1)	[13], [18], [26]–[29], [33]–[36], [82]–[109]	[19], [110], [111]	[112], [113]	[12], [114]–[116]	[71], [117]–[121]
Sensor fault and false association (A1)	[64], [92], [122]–[125]	[126]	[2]	[8], [9], [61], [65], [68], [114], [127], [128]	[72], [74], [76], [77], [129]–[134]
Terrain and wheel-ground contact (A2)	[27], [28], [30], [34], [37], [84], [99], [101], [102], [105], [107], [122], [135]–[139]	[6], [7], [14], [15], [19], [38]–[41], [47], [48], [140]–[148]	[55], [57], [112], [149]	[12], [22], [150]	[71], [75], [134], [151], [152]
Moving agents (A3)	[26], [32], [34], [86], [91], [97], [108], [137], [153]–[160]	[20], [39], [42]–[44], [46], [111], [141], [161]–[175]	[21], [49], [56], [58], [176]–[179]	–	[24], [73], [79], [152], [180]–[183]
Map and scene change (A3)	[26], [31], [64], [89], [95], [98], [108], [124], [125], [137], [155], [158], [160], [184]–[186]	[172], [187]	[5], [51], [113], [177], [188]–[190]	[62]	[24], [73], [81], [191]
Learned and semantic inference error (A4)	[37], [99], [107], [192]	[7], [45], [47], [140], [162], [166], [170], [173]–[175], [193]–[196]	[2], [3], [58], [149], [179], [197], [198]	–	[78], [80], [81], [119], [152], [180], [182], [183], [199]
Communication loss, delay and bandwidth (A5)	[64], [67]	[42], [126], [171]	–	[8]–[12], [59], [62], [63], [65], [66], [68]–[70], [116], [128], [150], [200]–[213]	[77], [214]
Actuator fault and saturation (A6)	–	[168], [215]–[218]	[2]	[69], [200], [219]–[221]	[72], [77], [132], [133], [222]–[225]
Computation and energy limits (A6)	[106], [137], [156], [186]	[42], [126], [175], [193]	[5], [21], [53]–[56], [113], [179], [188]	[62], [150], [201], [203], [206], [208]	[81], [117], [120], [214], [226]–[228]

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TABLE S11
SENSITIVITY OF THE SHARE OF TECHNICAL WORKS THAT REPORT A NAMED DEGRADATION CONDITION AND A TASK OUTCOME IN THE SAME EXPERIMENT. ROWS VARY THE DEFINITION, THE CODING UNIT OR THE SUBSET. RF1-RF4 STATE WHETHER EACH FINDING FIXED IN CODEBOOK VERSION 1.0 BEFORE CODING HOLDS.

Row	Works	%	IV %	V %	VI %	VII %	VIII %	RF1	RF2	RF3	RF4
S-def-headline	81/244	33	1	64	47	24	40	yes	no	no	yes
S-def-D1	27/244	11	0	21	17	11	10	yes	no	no	yes
S-def-D1-nominal	11/244	5	0	5	11	7	2	yes	no	no	yes
S-def-physical	43/244	18	1	36	36	4	18	yes	no	no	yes
S-def-with-incidental	88/244	36	1	66	53	31	42	yes	no	no	yes
S-unit-any	83/244	34	1	66	47	24	42	yes	no	no	yes
S-card-round3	61/244	25	3	36	25	29	42	yes	yes	no	yes
S-depth-full	7/23	30	0	80	67	0	20	tie	no	no	yes
S-depth-light-legacy	33/86	38	0	62	58	40	22	yes	no	no	yes
S-depth-light-targeted	41/135	30	3	64	38	19	50	yes	no	no	yes
S-depth-no-legacy	48/158	30	2	67	42	17	45	yes	no	no	yes
Without Initial sample	63/199	32	2	61	42	28	47	yes	no	no	yes
Without Complementary publications	79/240	33	1	64	47	23	41	yes	no	no	yes
Without Thematic sample	61/184	33	2	67	46	18	39	yes	no	no	yes
Without Integration questions	76/229	33	2	64	47	20	43	yes	no	no	yes
Without Targeted IEEE coverage	75/224	33	2	63	49	26	41	yes	no	no	yes
Without Further targeted coverage	68/196	35	2	67	50	31	29	yes	no	no	yes
Without Recent IEEE coverage	68/208	33	0	63	47	24	38	yes	no	no	yes
Without Coverage additions	77/228	34	2	65	50	28	40	yes	no	no	yes
S-year-2021-2023	10/33	30	0	60	50	0	40	tie	no	no	yes
S-year-2024	11/38	29	0	80	17	17	17	yes	no	no	yes
S-year-2025	40/87	46	0	64	60	50	54	yes	no	no	yes
S-year-2026	20/86	23	3	57	45	26	0	no	no	no	yes
S-quota	33/96	34	3	61	50	39	46	yes	no	no	yes
S-non-quota	48/148	32	0	66	45	15	37	yes	no	no	yes
S-quota-theme-T1	0/16	0	0	–	–	–	–	n/a	n/a	n/a	yes
S-quota-theme-T2	2/16	12	8	33	–	–	–	n/a	n/a	n/a	no
S-quota-theme-T3	10/16	62	–	67	–	0	–	n/a	n/a	n/a	n/a
S-quota-theme-T4	6/16	38	–	–	–	38	–	n/a	n/a	n/a	n/a
S-quota-theme-T5	8/16	50	–	–	50	–	–	n/a	n/a	n/a	n/a
S-quota-theme-T6	7/16	44	0	–	–	100	46	n/a	n/a	n/a	n/a
S-no-ieee-only	62/188	33	0	62	48	25	38	yes	no	no	yes
S-no-coverage	77/228	34	2	65	50	28	40	yes	no	no	yes
S-platform-WHL	69/172	40	3	64	53	25	40	yes	no	no	yes
S-platform-WHL-mixed	69/182	38	3	64	53	24	40	yes	no	no	yes
S-platform-non-WHL	12/72	17	0	67	17	23	40	yes	no	no	yes

Definition rows change the measure and use all technical works. Other rows apply the headline measure to a subset. A finding is n/a when a function it compares has fewer than three works in the subset. All shares describe the purposively selected sample.

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TABLE S12
AGREEMENT BETWEEN THE INDEPENDENT SECOND CODING AND THE FROZEN PRIMARY CODES ON THE 40 SAMPLED WORKS, BEFORE ADJUDICATION. BOTH CODERS ARE LANGUAGE-MODEL SESSIONS CONFIGURED FOR DIFFERENT MODELS. THE VALUES THEREFORE DESCRIBE HOW CONSISTENTLY TWO SUCH CODERS APPLY THE CODEBOOK TO SHARED EVIDENCE. THEY ARE NOT HUMAN INTER-RATER RELIABILITY.

Attribute	Unit	n	Cat.	Agr. %	κ	Exact %	Flag κ	Works	P/S/B
Functions	non-principal function flags, pooled	160	2	96.2	0.89	85.0	0.75–1.00	6	2/4/0
Addressed sources	source flags, pooled	360	2	96.4	0.86	70.0	0.66–1.00	12	4/8/0
Manipulated sources (D1)	source flags, pooled	360	2	94.7	0.53	67.5	–	13	4/8/1
Selected sources (D2)	source flags, pooled	360	2	95.8	0.61	70.0	–	12	4/7/1
Incidental sources (D3)	source flags, pooled	360	2	100.0	1.00	100.0	–	0	0/0/0
Named sources (D1 or D2)	source flags, pooled	360	2	97.2	0.86	80.0	0.63–1.00	8	–
Response family	primary family	40	20	85.0	0.84	–	–	6	–
Response family	full value, with the optional secondary family of version 1.0	40	46	30.0	0.29	–	–	28	24/0/4
Publication class	strongest class	40	3	72.5	0.58	–	–	11	–
Nominal control	value	40	3	67.5	0.42	–	–	13	4/9/0
Single-level injection	value	40	2	92.5	0.54	–	–	3	2/1/0
Task setting	value	40	4	90.0	0.86	–	–	4	1/2/1
Same-experiment class	value	40	3	82.5	0.69	–	–	7	4/3/0
Same-experiment setting	value	40	3	92.5	0.86	–	–	3	1/2/0
Degradation evidence level	value	40	7	85.0	0.81	–	–	6	3/2/1
Drivetrain effect	value	40	2	87.5	−0.04	–	–	5	5/0/0

n counts the scored units and Cat. the categories that either coder used. κ is unweighted Cohen's kappa (n/a when both coders use one category). Set-valued attributes are scored as one binary flag per value. For them, κ and Agr. pool all flag decisions of the attribute (for functions the four non-principal functions of each work). Exact gives the share of works with identical sets. Flag κ gives the range of the per-flag kappas (functions over the 32 works whose principal function differs, sources over all 40 works). The unit of response family fixed in codebook version 1.0 is the primary family. The second row compares the full value, which in codebook version 1.0 could add an optional secondary family. Named sources and publication class are derived from the three class columns. Works counts the works with differing values and P/S/B how the logged disagreements were resolved: primary kept, second adopted, both revised.

TABLE S13
FUNCTION-PAIR CO-OCCURRENCE WITH EVIDENCE LEVELS. NAMED, D1 AND PHYSICAL COUNT WORKS WITH A SAME-EXPERIMENT COMBINATION OF A NAMED CONDITION AND A TASK OUTCOME, A D1 CONDITION, OR A PHYSICAL ROBOT. WHEELED JOINT COUNTS SUCH PHYSICAL COMBINATIONS ON TARGET WHEELED ROBOTS.

Pair	Works	Named	D1	Physical	Wheeled joint	Wheeled	Level
LOC–PLN	14	5	3	1	1	12	P
LOC–SEM	8	2	2	2	2	5	P
LOC–MRC	25	1	1	0	0	18	P
LOC–REL	19	2	0	1	1	11	P
PLN–SEM	20	14	3	10	9	18	P
PLN–MRC	23	12	6	3	2	20	P
PLN–REL	25	15	2	9	7	19	P
SEM–MRC	1	0	0	0	0	0	none
SEM–REL	1	1	0	0	0	0	R
MRC–REL	12	5	5	0	0	6	P

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TABLE S14
INTERFACE CANDIDATES RANKED BY THE RULE FIXED IN CODEBOOK VERSION 1.0 BEFORE CODING. A CANDIDATE IS A FUNCTION PAIR WITH ONE COUPLING SOURCE. N COUNTS ITS WORKS, J THE WORKS WITH DIRECT JOINT WHEELED EVIDENCE AND W THE TARGET WHEELED WORKS. RANK A DROPS THE INTEGRATION-QUESTIONS STAGE AND RANK B SORTS BY GAP SHARE.

Rank	Candidate	n	j	w	Gap	Rank A	Rank B	Status	Reason
1	LOC-MRC:COM	17	0	12	17	1	12	Interface 4	
2	PLN-MRC:COM	15	1	12	14	2	16	rejected	Merged into Interface 4. Rendezvous, reception-aware control and communication-aware allocation already plan around links (Sec. VII-C). The open part is acceptance of information after reconnection.
3	LOC-MRC:SFA	9	0	6	9	3	13	rejected	Merged into Interface 4. Outlier loop closures and faulty shared landmarks are the corrupted content that its comparison injects as landmark bias.
4	LOC-REL:PER	9	1	5	8	4	17	Interface 1	
5	LOC-REL:SFA	7	0	3	7	7	5	Interface 1	
6	PLN-SEM:DYN	10	4	8	6	6	25	rejected	Merged into Interface 2, whose comparison varies obstacle-track quality. Four works already give direct joint wheeled evidence for social or semantic choices among moving people.
7	MRC-REL:SFA	6	0	3	6	13	7	rejected	Merged into Interface 4. Fault propagation through shared state is one of its primary outcomes.
8	LOC-MRC:PER	6	0	5	6	5	14	rejected	Merged into Interface 4. Teammate ranges and maps serve as complementary sensing (Sec. IV-B), and their acceptance after a partition is the interface question.
9	PLN-REL:ACT	5	0	5	5	12	1	rejected	Named as a further gap in Sec. X-E. The five works accommodate modeled actuator faults inside one tracking controller, without a response chosen across functions.
10	PLN-SEM:MOD	7	3	6	4	8	26	Interface 2	
11	LOC-PLN:PER	5	1	5	4	9	18	Interface 3	
26	PLN-SEM:RES	6	4	6	2	29	29	Interface 2	
–	LOC-PLN:TER	2	0	1	2	–	–	Interface 3	
–	SEM-REL:MOD	1	0	0	1	–	–	Interface 2	
–	SEM-REL:RES	0	0	0	0	–	–	Interface 2	

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